HOMEWORK 07

Sensor Systems

Due date

Squad A: Thursday, November 5, 14:30
Squad B: Sunday, November 8, 14:30

7a. ADC scan using DMA

The objective of this project is to acquire 3 voltages (potentiometer, temperature sensor, Vref) every 1 second and send them to a remote terminal. The acquisitions are started via hardware by a timer, and the data are saved in the microcontroller memory using DMA.

7b. Light Dependent Resistor (LDR)

The objective of this project is:

Step 1: acquire the LDR resistance value every 1 ms and send its average value to a remote terminal every 1 second.

Step 2: convert the resistance value to a lux level and send it to the remote terminal.

Hint: *make use of the half complete callback.*